

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250275729

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Smith; Michael Eric et al.

JOINT FOR A SCAN WINDOW FORMED TO A CYLINDER

Abstract

A scan window for an imaging system includes at least one flexible sheet made of an X-ray transparent material and having a first end and a second end. The scan window also includes at least one tapered joint formed between the first end and the second end and secured via a bonding agent to form a joined member with the at least one flexible sheet.

Inventors: Smith; Michael Eric (Whitewater, WI), Smith; Brandon Allan (Waukesha, WI), Heiman; Wesley George (Mukwonago, WI)

Applicant: GE Precision Healthcare LLC (Waukesha, WI)

Family ID: 85175805

Appl. No.: 19/210542

Filed: May 16, 2025

Related U.S. Application Data

parent US division 17678525 20220223 parent-grant-document US 12324692 child US 19210542

Publication Classification

Int. Cl.: A61B6/03 (20060101); A61B6/00 (20240101); B29C65/00 (20060101); G01N23/04 (20180101); G01V5/22 (20240101)

U.S. Cl.:

CPC A61B6/035 (20130101); A61B6/032 (20130101); A61B6/4435 (20130101); B29C66/4324 (20130101); G01N23/04 (20130101); G01N2223/317 (20130101); G01V5/22 (20240101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. application Ser. No. 17/678,525, filed on Feb. 23, 2022, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] The subject matter disclosed herein relates to imaging systems and, more particularly, to utilizing a joint for forming a scan window into a cylinder for utilization in an imaging system.

[0003] In computed tomography (CT), X-ray radiation spans an object or a subject of interest being scanned, such as a human patient, baggage, or other object, and a portion of the radiation impacts a detector where the image data is collected. In digital X-ray systems a photodetector produces signals representative of the amount or intensity of radiation impacting discrete pixel regions of a detector surface. The signals may then be processed to generate an image that may be displayed for review. In the images produced by such systems, it may be possible to identify and examine the internal structures and organs within a subject's body. In CT imaging systems a detector array, including a series of detector elements or sensors, produces similar signals through various positions as a gantry is displaced around a subject or object being imaged, allowing volumetric image reconstructions to be obtained.

[0004] A CT imaging system may include an annular scan window disposed within an interior wall of the gantry formed within an opening or a bore of the gantry. The scan window is made of an X-ray transparent material that enables X-rays emitted from an X-ray source to pass through an object or subject being imaged for detection by a detector. Typically, two ends of a sheet made of the X-ray transparent material are joined together via joint to form a joined member. Larger scan windows are difficult to manage due to cracking, manufacturing of the joint, and contamination. Over time, the joint can crack. In addition, the joint can become contaminated during the cracking process, prior to actual mechanical failure.

BRIEF DESCRIPTION

[0005] Certain embodiments commensurate in scope with the originally claimed subject matter are summarized below. These embodiments are not intended to limit the scope of the claimed subject matter, but rather these embodiments are intended only to provide a brief summary of possible forms of the subject matter. Indeed, the subject matter may encompass a variety of forms that may be similar to or different from the embodiments set forth below.

[0006] In one embodiment, a scan window for an imaging system is provided. The scan window includes at least one flexible sheet made of an X-ray transparent material and having a first end and a second end. The scan window also includes at least one tapered joint formed between the first end and the second end and secured via a bonding agent to form a joined member with the at least one flexible sheet.

[0007] In another embodiment, a gantry housing for an imaging system is provided. The gantry housing includes a scan window. The scan window includes at least one sheet made of an X-ray transparent material and having a first tapered end and a second tapered end. The scan window also includes at least one tapered joint formed between the first tapered end and the second tapered end and secured via a bonding agent to form a joined member with the sheet.

[0008] In a further embodiment, a joined member is provided. The joined member includes a flexible sheet made of a plastic material and having a first tapered end and a second tapered end. The flexible cylinder member also includes a tapered joint formed between the first tapered end and the second tapered end overlapping with each other and secured via a bonding agent to form the joined member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] These and other features, aspects, and advantages of the disclosed subject matter will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:

[0010] FIG. 1 is a combined pictorial view and block diagram of a computed tomography (CT) imaging system as discussed herein;

[0011] FIG. 2 is a perspective view of a gantry having an opening or bore extending therethrough, in accordance with aspects of the present disclosure;

[0012] FIG. 3 is a perspective view of a main body of a joined member (e.g., having an angled joint), in accordance with aspects of the present disclosure;

[0013] FIG. 4 is a perspective view of a main body of a joined member (e.g., having a parallel joint), in accordance with aspects of the present disclosure;

[0014] FIG. 5 is a top view of a sheet for forming the main body of the joined member in FIG. 3, in accordance with aspects of the present disclosure;

[0015] FIG. 6 is a top view of a sheet for forming the main body of the joined member in FIG. 4, in accordance with aspects of the present disclosure;

[0016] FIG. 7 is an end view of a portion of a main body of a joined member having a tapered joint (e.g., having a full ramp on both sides), in accordance with aspects of the present disclosure;

[0017] FIG. 8 is an end view of a portion of a main body of a joined member having a tapered joint (e.g., having a step on both sides), in accordance with aspects of the present disclosure;

[0018] FIG. 9 is an end view of a portion of a main body of a joined member having a tapered joint (e.g., having a full ramp on one side and a step on one side), in accordance with aspects of the present disclosure;

[0019] FIG. 10 is an end view of a portion of a main body of a joined member having a tapered joint (e.g., having mating radii on both sides), in accordance with aspects of the present disclosure;

[0020] FIG. 11 is a perspective view of a main body of a joined member (e.g., formed from multiple sheets), in accordance with aspects of the present disclosure;

[0021] FIG. 12 is a top view of a plurality of sheets for forming the main body of the joined member in FIG. 11, in accordance with aspects of the present disclosure; and

[0022] FIG. 13 is a perspective view of a cylindrical member, in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

[0023] One or more specific embodiments will be described below. In an effort to provide a concise description of these embodiments, not all features of an actual implementation are described in the specification. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be appreciated that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

[0024] When introducing elements of various embodiments of the present subject matter, the articles “a,” “an,” “the,” and “said” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Furthermore, any numerical examples in the following discussion are intended to be non-limiting, and thus additional numerical

values, ranges, and percentages are within the scope of the disclosed embodiments.

[0025] While aspects of the following discussion may be provided in the context of medical or health care imaging, it should be appreciated that the present techniques are not limited to such medical contexts. Indeed, the provision of examples and explanations in such a medical context is only to facilitate explanation by providing instances of real-world implementations and applications. However, the present approaches may also be utilized in other contexts, such as tomographic image reconstruction for industrial CT used in non-destructive inspection of manufactured parts or goods (i.e., quality control or quality review applications), and/or the non-invasive inspection of packages, boxes, luggage, and so forth (i.e., security or screening applications). In general, the present approaches may be useful in any imaging or screening context utilizing a generally annular or cylindrical scan window.

[0026] The present disclosure provides embodiments for an annular or cylindrical scan window. In particular, the annular or cylindrical scan window is disposed in a gantry of an imaging system (e.g., CT imaging system). In certain embodiments, the annular or cylindrical scan window includes at least one sheet having ends joined by an engineered joint. For example, the annular or cylindrical scan window includes at least one sheet having the ends (e.g., tapered ends) joined (e.g., fastened, bonded or secured) together via a bonding agent to form a tapered joint. The tapered ends overlap with each other to form the tapered joint. The forming of the tapered joint enables the at least one sheet to form the annular or cylindrical shape. In certain embodiments, more than one sheet may be joined together via tapered joints to form the annular or cylindrical scan window. The tapered ends at the beginning of the taper each include a thickness that is at least half a thickness of the sheet outside the tapered joint, thus, increasing the strength of the joint where the ends of the sheet are joined and extending the life of the scan window. The tapered joint is tapered over a distance that is 5 to 20 times the thickness of the sheet, thus, spreading out stresses as the annular or cylindrical scan window is being flexed (e.g., during removal or insertion into the gantry) compared to other joint configurations. The disclosed embodiments provide longer flexing life and less contamination risk for the annular or cylindrical scan window. In addition, the disclosed embodiments enable manufacturing of the scan window from commonly available plastic materials (e.g., thermoplastic polymers, polycarbonates). Further, the disclosed embodiments may also be utilized outside of medical or imaging contexts to form flexible joined members with an engineered joint.

[0027] With the preceding in mind and referring to FIG. 1, a computed tomography (CT) imaging system **10** is shown, by way of example. The CT imaging system includes a gantry **12**. The gantry **12** has an X-ray source **14** that projects a beam of X-rays **16** toward a detector assembly **15** on the opposite side of the gantry **12**. The X-ray source **14** projects the beam of X-rays **16** through a pre-patient collimator assembly **13** that determines the size and shape of the beam of X-rays **16**. The detector assembly **15** includes a collimator assembly **18** (a post-patient collimator assembly), a plurality of detector modules **20** (e.g., detector elements or sensors), and data acquisition systems (DAS) **32**. The plurality of detector modules **20** detect the projected X-rays that pass through a subject or object **22** being imaged, and DAS **32** converts the data into digital signals for subsequent processing. Each detector module **20** in a conventional system produces an analog electrical signal that represents the intensity of an incident X-ray beam and hence the attenuated beam as it passes through the subject or object **22**. During a scan to acquire X-ray projection data, gantry **12** and the components mounted thereon rotate about a center of rotation **25** (e.g., isocenter) so as to collect attenuation data from a plurality of view angles relative to the imaged volume.

[0028] Rotation of gantry **12** and the operation of X-ray source **14** are governed by a control system **26** of CT imaging system **10**. Control system **26** includes an X-ray controller **28** that provides power and timing signals to an X-ray source **14**, a collimator controller **29** that controls a length and a width of an aperture of the pre-patient collimator **13** (and, thus, the size and shape of the beam of X-rays **16**), and a gantry motor controller **30** that controls the rotational speed and

position of gantry **12**. An image reconstructor **34** receives sampled and digitized X-ray data from DAS **32** and performs high-speed image reconstruction. The reconstructed image is applied as an input to a computer **36**, which stores the image in a storage device **38**. Computer **36** also receives commands and scanning parameters from an operator via console **40**. An associated display **42** allows the operator to observe the reconstructed image and other data from computer **36**. The operator supplied commands and parameters are used by computer **36** to provide control signals and information to DAS **32**, X-ray controller **28**, collimator controller **29**, and gantry motor controller **30**. In addition, computer **36** operates a table motor controller **44**, which controls a motorized table **46** to position subject **22** and gantry **12**. Particularly, table **46** moves portions of subject **22** through a gantry opening or bore **48**.

[0029] FIG. **2** is a perspective view of the gantry **12** having an opening or bore **48** extending through the gantry **12**. Although the following embodiments are discussed in terms of the CT imaging system, the embodiments may also be utilized with other imaging systems (e.g., positron emission tomography (PET) imaging system, CT/PET imaging system, single photon emission computed tomography (SPECT) imaging system, and other hybrid CT imaging systems, etc.) that include a scan window **50**. The gantry **12** includes a housing **52**, the gantry **12** includes a stationary section and a rotating section. The rotating section rotates around an axis which lies in the z-direction (which corresponds to a body-axis direction) relative to the stationary section. The X-ray source **14** and the detector assembly **15** are rotating elements of the rotating section. The housing **52** is constructed to cover the components disposed in the inside of the gantry **12**. Moreover, the housing **52** forms space for movement in which the rotating section is capable of moving about the axis in the z-direction in the space surrounded by the housing **52**.

[0030] As depicted, the housing **52** includes a front cover **54** and a rear cover **58**. In another exemplary embodiment, the housing **52** may include a main cover (not shown) positioned between the front cover **54** and the rear cover **58**. The front cover **54** is fixed on the side of a front surface of the gantry **12**, while the rear cover **58** is fixed on the side of a back surface of the gantry **12**. The front cover **54** and a rear cover **58** are stationary. The number and structure of covers forming the housing **52** may differ from those depicted in FIG. **2**.

[0031] Within the bore **48**, a gap may be formed between the front cover **54** and the rear cover **58**. The scan window **50** is constructed so as to be fitted between the front cover **54** and the rear cover **58** to fill the gap. A worker (e.g., field engineer or employee in a manufacturing facility of an imaging system) can manually fit the scan window **50** within the bore **48** to fill the gap. The worker can also manually remove the scan window **50** from the bore **48**. The scan window **50** can be inserted or removed from the bore **48** flexing a portion of the scan window **50** inward to partially fold or deform the scan window **50**. The flexibility of the material used in the scan window **50** allows it to be installed or removed while the front cover **54** and rear cover **58** remain in place.

[0032] During an imaging session, a subject or patient is moved within the bore **48**. The scan window **50** self-supports and acts as a safety barrier to keep the subject or patient from contacting rotating components within the gantry **12**.

[0033] The scan window **50** includes a main body **60** and end members **62**, **64** (e.g., elastic members, flexible members, gaskets, etc.). The end members **62**, **64** are disposed on an outer surface of the main body **60** at ends **66**, **68**, respectively. The ends **66**, **68** are located adjacent to the front cover **54** and the rear cover **58**. The scan window **50** has a cylindrical or annular shape. The scan window **50** is open at both ends **66**, **68**.

[0034] The main body **60** is formed of a single sheet **70** made of a plastic material that is transparent to X-rays (e.g., having minimal attenuation) and is flexible or deformable. For example, the plastic material may be a thermoplastic polymer or a polycarbonate. The single sheet **70** may be made of another type of plastic material that is also transparent to X-rays and flexible or deformable.

[0035] As described in greater detail below, ends (e.g., tapered ends) of the single sheet **70** are

joined together via an engineered joint (e.g., a tapered joint) to form the cylindrical or annular shape. A length (e.g., width) of the engineered joint (e.g., in a direction perpendicular to a central axis of the scan window **50**) is approximately 5 to 20 times the thickness of the sheet **70**. In certain embodiments, the length (e.g., width) of the joint is approximately 10 times the thickness of the sheet **70**. The tapering of the joint spreads out the stresses as the joint is flexed.

[0036] The ends of single sheet **70** are secured together to form the engineered joint via a bonding agent. In certain embodiments, the bonding agent may be plastic-dissolving, UV-cured, or epoxy-based bonding agent. The bonding agent is utilized to fill in all potential empty volume in the engineered joint to provide continuous flow of the stresses through the joint during operation and flexing.

[0037] In certain embodiments, the scan window **50** may be formed of multiple sheets **70** joined together via multiple engineered joints (e.g., tapered joints). For example, an end (e.g., tapered end) of one sheet **70** may be secured to an end (e.g., tapered end) of another sheet **70** to form the engineered joint.

[0038] The engineered joint has the strength capability of the base material of the sheet **70**. The engineered joint has an increased life of at least 3 to 4 times the life of a typical joint used to form the main body **60** of the scan window **50**. The engineered joint may improve the IFR90 (infantile failure rate in the first 90 days) of the scan window **50**. The engineered joint may also improve the long-term reliability of the scan window **50** over a period of years. The engineered joint may be especially useful for large cover scanners (70 centimeters (cm) and above). Due to the reduced cracking risk compared with other joints, the engineered joint will receive less contaminants over time (which is a problem for larger scan windows with lots of surface area). The engineered joint also reduces downtime for the imaging system due to the longer window service life with flexing. In addition, the engineered joint enables simpler cleaning of the scan window **50** since it takes longer for the scan window **50** to crack. Further, the engineered joint maintains a more robust physical barrier between the subject or patient being scanned and internal parts of the gantry **12**.

[0039] FIG. **3** is a perspective view of a main body **60** of a joined member **50**, such as a scan window, having an angled joint. The main body **60** includes ends (e.g., tapered ends) of a single sheet **70** joined together to form an engineered joint **72** (e.g., tapered joint). The tapered joint **72** extends from the end **66** to the end **68**. As depicted, the tapered joint **72** is angled (i.e., non-parallel) relative to a central axis **74**. In certain embodiments (see FIG. **4**), the tapered joint **72** is parallel with the central axis **74**. FIG. **4** is a perspective view of a main body **60** of a joined member **50**, such as a scan window, having a joint **72** that is parallel with the central axis **74**. A length **76** (e.g., width) of the tapered joint **72** (e.g., in a direction perpendicular to the central axis **74** of the scan window **50**) is approximately 5 to 20 times a length **78** (e.g., thickness) of the sheet **70**. In certain embodiments, the width **76** of the joint is approximately **10** times the thickness **78** of the sheet **70**. The tapering of the joint **72** spreads out the stresses as the tapered joint **72** is flexed.

[0040] FIGS. **5** and **6** are top views of the sheets **70** for forming the respective main bodies **60** of the joined members **50**, such as scan windows in FIGS. **3** and **4**. The sheet **70** includes ends **80**, **82** (e.g., longitudinal ends) flanking a central portion **84** of the sheet **70**. Each end **80**, **82** is a tapered end. The ends **80**, **82** are joined together to form a tapered joint that is secured via a bonding agent. The ends **80**, **82** overlap to form the tapered joint. The end **80** tapers in a direction **86** (e.g., oriented from the end **82** to the end **80**). The end **82** tapers in a direction **88** (e.g., oriented from the end **80** to the end **82**). In certain embodiments, the ends **80**, **82** continuously taper. In certain embodiments, the ends **80**, **82** initially taper and then maintain a constant thickness toward a most distal location of the ends **80**, **82**.

[0041] The sheet **70** has a length **90** (e.g., longitudinal length) and a length **92** (e.g., width) (in the central portion **84** and perpendicular to the length **90**). The length **90** is greater than the length **92** (e.g., width). In FIG. **5**, a length **94** of the ends **80**, **82** is greater than the length **92** (e.g., width) of the central portion **84** of the sheet **70**. In FIG. **6**, the length **94** of the ends **80**, **82** is the same as the

length **92** (e.g., width) of the central portion **84** of the sheet **70**. In FIG. 5, the ends **80**, **82** are angled relative to the length **90**. In certain embodiments, an angle **96** formed between the ends **80**, **82** and the length **92** (e.g., width) may be approximately 60 degrees. In certain embodiments, the angle **96** may range from approximately 90 degrees (as depicted in FIG. 6) to approximately 30 degrees. The length **90** of the sheet **70** is approximately equal to the circumference of the bore **48**. In an exemplary embodiment, the design of the joined member or scan window **50** is best suited for sheet lengths **90** greater than approximately 6 inches with no upper limit. A flexible, short scan window of less than 6 inches in length can often self-support and may not need this type of engineered joint. However, as the sheet lengths increase, the need for this type of engineered joint is more critical.

[0042] The sheet **70** is made of a plastic material that is transparent to X-rays (e.g., having minimal attenuation) and is flexible or deformable. For example, the plastic material may be a thermoplastic polymer or polycarbonate. The single sheet **70** may be made of another plastic material that is also transparent to X-rays and deformable. In certain embodiments, where a flexible cylindrical member is made utilizing a tapered joint for non-imaging applications, the sheet **70** made of a plastic material that is flexible or deformable (without having to take into account the material's transparency to X-rays) and has the desired material strength for the purpose of the flexible cylindrical member.

[0043] FIG. 7 is an end view of a portion of the main body **60** of a joined member **50**, such as a scan window having a tapered joint **72** (e.g., having a full ramp on both sides). A sheet **70** has ends **80**, **82** that overlap to form the tapered joint **72**. More specifically, the ends **80**, **82** taper and overlap with each other to form the tapered joint **72**. Each tapered end **80**, **82** has a full ramp. In other words (as shown with respect to tapered end **80**), each tapered end **80**, **82** decreases (**80**) or increases (with respect to tapered end **82**) in thickness **78** from a beginning location **98** (e.g., upstream location) of the taper to an ending location **100** (e.g., downstream location) at the most distal location of the ends **80**, **82**. As depicted in FIG. 7, each tapered end **80**, **82** continuously decreases or narrows in thickness **78** along the length **76** of the tapered joint **72** from the beginning location **98** to the ending location **100** at the most distal location of the ends **80**, **82**. The most distal location of the tapered ends **80**, **82** forms a mating point. The thickness **78** of the tapered ends **80**, **82** at the beginning of the taper is approximately the same as the thickness **78** of the sheet **70** in a location outside of the tapered joint **72**. In certain embodiments, the thickness **78** may be approximately 1.5 millimeters (mm) (± 0.05 mm) or less. In certain embodiments, the thickness **78** may be approximately 1.0 mm (± 0.05 mm) or less. Utilization of the tapered joint **72** enables the thickness **78** of the sheet **70** to be less than 1.0 mm (± 0.05 mm). In addition, the thickness **78** at the beginning location **98** of the taper will cause it to take longer for a crack to develop and progress under flexion of the main body **60** compared to other types of joints.

[0044] The length **76** (e.g., width) of the tapered joint **72** (e.g., in a direction perpendicular to the central axis **74** of the scan window **50**) is approximately 5 to 20 times a length **78** (e.g., thickness) of the sheet **70**. In certain embodiments, the length **76** (e.g., width) of the joint **72** (e.g., in a direction **101**, which is perpendicular to the central axis of the scan window **50**) is approximately 10 times the thickness **78** of the sheet **70**. The tapering of the joint **72** spreads out the stresses as the tapered joint **72** is flexed.

[0045] Bonding agent is disposed along an interface **102** between the tapered ends **80**, **82**. Due to the bonding agent, the thickness **78** of the main body **60** in the region along the tapered joint **72** may be greater than the thickness **78** of the sheet **70** at a location outside the tapered joint **72**. An interior surface of the main body **60** at location **100** may be polished to remove excess bonding agent, while on exterior surface of the main body **60** at location **100** any excess bonding agent may remain.

[0046] FIG. 8 is an end view of a portion of a main body **60** of a joined member **50**, such as a scan window having a tapered joint **72** (e.g., having steps on both sides). A sheet **70** has ends **80**, **82** that

overlap to form the tapered joint 72. More specifically, the ends 80, 82 taper and overlap with each other to form the tapered joint 72. Each tapered end 80, 82 has a partial ramp that ends in a step 104 (e.g., mating step). In other words, each tapered end 80, 82 decreases in thickness 78 from a beginning location 98 (e.g., upstream location) of the taper to an ending location 100 (e.g., downstream location) at the most distal location of the ends 80, 82. As depicted in FIG. 8, each tapered end 80, 82 continuously decreases or narrows in the thickness 78 along the length 76 of the tapered joint 72 from the beginning location 98 to the step 104. The step 104 has a length 106 (e.g., thickness) of approximately 0.25 mm (± 0.05 mm). The thickness 78 of the tapered ends 80, 82 at the beginning of the taper is approximately the same as the thickness 78 of the sheet in a location outside of the tapered joint 72. In certain embodiments, the thickness 78 may be approximately 1.5 millimeters (mm) (± 0.05 mm) or less. In certain embodiments, the thickness 78 may be approximately 1.0 mm (± 0.05 mm) or less. Utilization of the tapered joint 72 enables the thickness 78 of the sheet 70 to be less than 1.0 mm (± 0.05 mm). In addition, the thickness 78 at the beginning location 98 of the taper will cause it to take longer for a crack to develop and progress under flexion of the main body 60 compared to other types of joints.

[0047] The length 76 (e.g., width) of the tapered joint 72 (e.g., in a direction perpendicular to the central axis 74 of the scan window 50) is approximately 5 to 20 times a length 78 (e.g., thickness) of the sheet 70. In certain embodiments, the length 76 (e.g., width) of the joint is approximately 10 times the thickness 78 of the sheet 70. The tapering of the joint 72 spreads out the stresses as the tapered joint 72 is flexed.

[0048] Bonding agent is disposed along the interface 102 between the tapered ends 80, 82. Due to the bonding agent, the thickness 78 of the main body 60 in the region along the tapered joint 72 may be greater than the thickness 78 of the sheet 70 at a location outside the tapered joint 72. An interior surface of the main body 60 at location 100 may be polished to remove excess bonding agent, while on exterior surface of the main body 60 at location 100 any excess bonding agent may remain.

[0049] FIG. 9 is an end view of a portion of a main body 60 of a joined member 50, such as a scan window having a tapered joint 72 (e.g., having a full ramp on one side and a step on one side). A sheet 70 has ends 80, 82 that overlap to form the tapered joint 72. More specifically, the ends 80, 82 taper and overlap with each other to form the tapered joint 72. One tapered end 80 has a full ramp that ends in a mating point. The other tapered end 82 has a partial ramp that ends in the step 104 (e.g., mating step 104). In other words, each tapered end 80, 82 decreases in thickness 78 from a beginning location 98 (e.g., upstream location) of the taper to an ending location 100 (e.g., downstream location) at the most distal location of the ends 80, 82. As depicted in FIG. 9, the tapered end 80 continuously decreases or narrows in the thickness 78 along the length 76 of the tapered joint 72 from the beginning location 98 to the step 104. The tapered end 82 continuously decreases or narrows in thickness 78 along the length 76 (e.g., width) of the tapered joint 72 from the beginning location 98 to the mating point. The step 104 has a length 106 (e.g., thickness) of approximately 0.25 mm (± 0.05 mm). The thickness 78 of the tapered ends 80, 82 at the beginning of the taper is approximately the same as the thickness 78 of the sheet 70 in a location outside of the tapered joint 72. In certain embodiments, the thickness 78 may be approximately 1.5 millimeters (mm) (± 0.05 mm) or less. In certain embodiments, the thickness 78 may be approximately 1.0 mm (± 0.05 mm) or less. Utilization of the tapered joint 72 enables the thickness 78 of the sheet 70 to be less than 1.0 mm (± 0.05 mm). In addition, the thickness 78 at the beginning location 98 of the taper will cause it to take longer for a crack to develop and progress under flexion of the main body 60 compared to other types of joints.

[0050] The length 76 (e.g., width) of the tapered joint 72 (e.g., in a direction perpendicular to the central axis 74 of the scan window 50) is approximately 5 to 20 times a length 78 (e.g., thickness) of the sheet 70. In certain embodiments, the length 76 (e.g., width) of the joint is approximately 10 times the thickness 78 of the sheet 70. The tapering of the joint 72 spreads out the stresses as the

tapered joint **72** is flexed.

[0051] Bonding agent is disposed along the interface **102** between the tapered ends **80, 82**. Due to the bonding agent, the thickness **78** of the main body **60** in the region along the tapered joint **72** may be greater than the thickness **78** of the sheet **70** at a location outside the tapered joint **72**. An interior surface of the main body **60** at location **100** may be polished to remove excess bonding agent, while on exterior surface of the main body **60** at location **100** any excess bonding agent may remain.

[0052] FIG. **10** is an end view of a portion of a main body **60** of a joined member **50**, such as a scan window having a tapered joint **72** (e.g., having mating radii on both sides). A sheet **70** has ends **80, 82** that may be joined together or overlap to form the tapered joint **72**. More specifically, the ends **80, 82** taper and overlap with each other to form the tapered joint **72**. Each tapered end **80, 82** ends in a radius **108** (e.g., rounded edge) that narrows in the thickness **78**. In particular, each tapered end **80, 82** initially decreases in thickness **78** from at a beginning location **98** (e.g., upstream location) of the taper. Then each tapered end **80, 82** has approximately a constant thickness **78** up until the radius **108**. Each tapered end **80, 82** beginning at the radius **108** continuously decreases in thickness **78** until an ending location **100** (e.g., downstream location) at the most distal location of the ends **80, 82**. The thickness **78** of the tapered ends **80, 82** at the beginning of the taper is approximately the same as the thickness **78** of the sheet **70** in a location outside of the tapered joint **72**. In certain embodiments, the thickness **78** may be approximately 1.5 millimeters (mm) (± 0.05 mm) or less. In certain embodiments, the thickness **78** may be approximately 1.0 mm (± 0.05 mm) or less. Utilization of the tapered joint **72** enables the thickness **78** of the sheet **70** to be less than 1.0 mm (± 0.05 mm). In addition, the thickness **78** at the beginning location **98** of the taper will cause it to take longer for a crack to develop and progress under flexion of the main body **60** compared to other types of joints.

[0053] The length **76** (e.g., width) of the tapered joint **72** (e.g., in a direction perpendicular to the central axis **74** of the scan window **50**) is approximately 5 to 20 times a length **78** (e.g., thickness) of the sheet **70**. In certain embodiments, the length **76** (e.g., width) of the joint is approximately 10 times the thickness **78** of the sheet **70**. The tapering of the joint **72** spreads out the stresses as the tapered joint **72** is flexed.

[0054] Bonding agent is disposed along the interface **102** between the tapered ends **80, 82**. Due to the bonding agent, the thickness **78** of the main body **60** in the region along the tapered joint **72** may be greater than the thickness **78** of the sheet **70** at a location outside the tapered joint **72**. An interior surface of the main body **60** at location **100** may be polished to remove excess bonding agent, while on exterior surface of the main body **60** at location **100** any excess bonding agent may remain.

[0055] FIG. **11** is a perspective view of a main body **60** of a joined member **50**, such as a scan window. The main body **60** of the joined member **50** is made of a plurality of sheets **70** (e.g., a first sheet **110** and a second sheet **112**, also depicted in FIG. **12**). FIG. **12** is a top view of a plurality of sheets **70** (e.g., a first sheet **110** and a second sheet **112**) for forming the main body **60** of the joined member **50** in FIG. **11**. As depicted in FIG. **11**, the plurality of sheets **70** (e.g., a first sheet **110** and a second sheet **112**) are coupled together via multiple tapered joints **72** (e.g., a first tapered joint **114** and a second tapered joint **116**.) As depicted in FIGS. **11** and **12**, each sheet **110, 112** has tapered ends **80, 82**. End **80** of the first sheet **110** and the end **82** of the second sheet **112** are joined together (and secured via a bonding agent) to form the first tapered joint **114**. End **80** of the sheet **110** and the end **82** of the second sheet **112** are jointed together (and secured via a bonding agent) to form the second tapered joint **116**. The tapered joints **114, 116** may be as described above. In certain embodiments, the number of sheets **70** and tapered joints to form the joined member **50** may be greater than two.

[0056] As noted above, the tapered joint may be utilized with any flexible member having a cylindrical shape. For example, FIG. **13** depicts a cylindrical member **118** having a tapered joint

120. The cylindrical member **118** includes a side wall **122** made of a flexible material. The cylindrical shape of the side wall **122** is formed by the joining of the ends of a sheet **122** and securing of the ends with bonding agent to form the tapered joint **120**. The tapered joint **120** may be as described above. The flexible material of the side wall **122** may be made of any plastic material having the desired flexibility and material strength to enable the cylindrical member **118** to be utilized as desired. The cylindrical member **118** includes a first end **124** and a second end **126**. One or both ends **124**, **126** of the cylindrical member **118** may be open or closed.

[0057] Technical effects of the disclosed embodiments include forming a main body of a scan window by coupling ends of a sheet together to form an engineered joint (e.g., tapered joint). The engineered joint has the strength capability of the base material of the sheet. The engineered joint has an increased life of at least 3 to 4 times the life of a typical joint used to form the main body of the scan window. The engineered joint may improve the IFR90 of the scan window. The engineered joint may also improve the long term reliability of the scan window over a period of years. Due to the reduced cracking risk compared with other joints, the engineered joint will receive less contaminants over time (which is a problem for larger scan windows with lots of surface area). The engineered joint also reduces downtime for the imaging system due to the longer window service life with flexing. In addition, the engineered joint enables simpler cleaning of the scan window since it takes longer for the scan window to crack. Further, the housing and scan window maintains a more robust physical barrier between the patient and internal parts of the gantry.

[0058] The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . . ” or “step for [perform]ing [a function] . . . ”, it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

[0059] This written description uses examples to disclose the present subject matter, including the best mode, and also to enable any person skilled in the art to practice the subject matter, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the subject matter is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A gantry housing for an imaging system, comprising: a scan window comprising: at least one sheet made of an X-ray transparent material and having a first tapered end and a second tapered end; and at least one tapered joint formed between the first tapered end and the second tapered end and secured via a bonding agent to form a joined member with the at least one sheet.
2. The gantry housing of claim 1, wherein a beginning of a taper of both the first tapered end and the second tapered end has a respective thickness that is at least half a thickness of the at least one sheet at a location outside of the at least one tapered joint.
3. A joined member, comprising: a flexible sheet made of a plastic material and having a first tapered end and a second tapered end; and a tapered joint formed between the first tapered end and the second tapered end overlapping with each other and secured via a bonding agent to form the joined member.
4. The gantry housing of claim 1, wherein the first tapered end and the second tapered end, respectively, that overlap with each other to form the at least one tapered joint.

5. The gantry housing of claim 4, wherein each of the first tapered end and the second tapered end of the at least one flexible sheet continuously narrows from a beginning of the respective end to a most distal location of the respective end.
 6. The gantry housing of claim 5, wherein the most distal location of at least one of the first tapered end and the second tapered end has a mating step.
 7. The gantry housing of claim 5, wherein the most distal location of both the first tapered end and the second tapered end has the mating step.
 8. The gantry housing of claim 5, wherein the most distal location of at least one of the first tapered end and the second tapered end has a mating point.
 9. The gantry housing of claim 5, wherein the most distal location of both the first tapered end and the second tapered end has the mating point.
 10. The gantry housing of claim 5, wherein the most distal location of the first tapered end has a mating step and the most distal location of the second tapered end has a mating point.
 11. The gantry housing of claim 4, wherein each of the first tapered end and the second tapered end of the at least one flexible sheet narrows at a beginning of the respective end, then has a constant thickness until near a most distal location of the respective end before continuously narrowing to the most distal location.
 12. The gantry housing of claim 11, wherein the most distal location of both the first end and the second end is rounded.
 13. The gantry housing of claim 2, wherein the beginning of the taper of both the first tapered end and the second tapered has a respective thickness that is approximately the same as the thickness of the at least one flexible sheet at the location outside of the at least one tapered joint.
 14. The gantry housing of claim 1, wherein the at least one tapered joint is parallel with a central axis of the scan window.
 15. The gantry housing of claim 1, wherein the at least one tapered joint is non-parallel with a central axis of the scan window.
 16. The gantry housing of claim 1, wherein a ratio of a respective length of both the first tapered end and the second tapered end in a direction perpendicular to a central axis of the scan window relative to a thickness of the at least one flexible sheet ranges from 5:1 to 20:1.
 17. The gantry housing of claim 16, wherein the ratio is 10:1.
-